

3. Beta thalassemia is a fairly common blood disorder worldwide. Beta thalassemia occurs frequently in people from Mediterranean and African countries. Normal homozygous individuals (**TT**) have normal red blood cells that are easily infected with the malarial parasite. Thus in the past many of these individuals became very ill from the parasite and died.

Individuals who are homozygous (**tt**) often die because of the complications of anaemia.

Individuals with the heterozygous condition (**Tt**) have mutant haemoglobin that means the malaria parasite cannot survive well in their red blood cells.

- (a) (i) The frequency of the **t** allele is 2% in a population.

Using the formulae given, calculate the **percentage** of the population that will be heterozygous (**Tt**) for the thalassemia gene. [3]

$$p + q = 1.0$$

$$p^2 + 2pq + q^2 = 1.0$$

Percentage of population that are heterozygous = .....

- (ii) Suggest the advantage of being a **Tt** individual in a region where malaria is present. [2]

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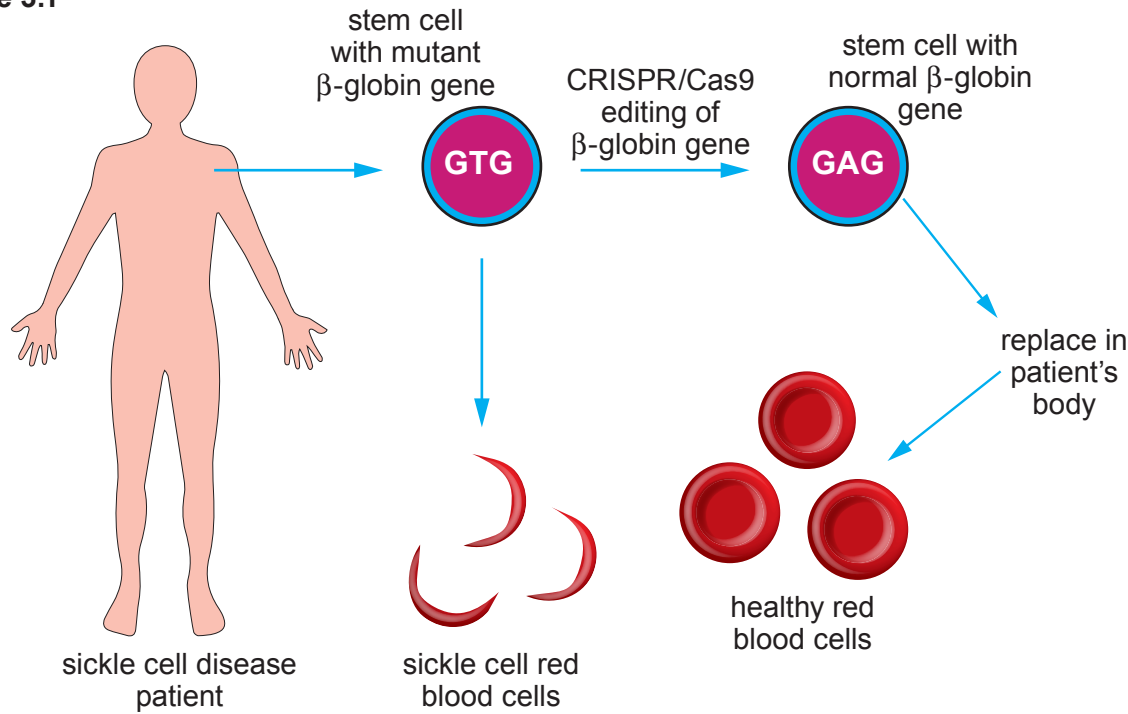
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A different form of anaemia is sickle cell disease (SCD). SCD is caused by a point mutation in the  $\beta$ -globin gene. A single nucleotide substitution from A to T in the codon for the sixth amino acid in the  $\beta$ -globin protein converts a glutamic acid to a valine that leads to the production of sickle haemoglobin.

- (b) One possible way of treating sickle cell disease is using gene therapy and stem cells as shown in **Image 3.1**. Stem cells are removed from the patient and treated with a gene editing technique called CRISPR/Cas9. The repaired gene can then function normally and the treated stem cells are replaced in the patient.

Image 3.1



- (i) State why stem cells must be used in this technique. [2]

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- (ii) Describe **two** advantages of using this type of stem cell treatment over using embryonic stem cells. [2]

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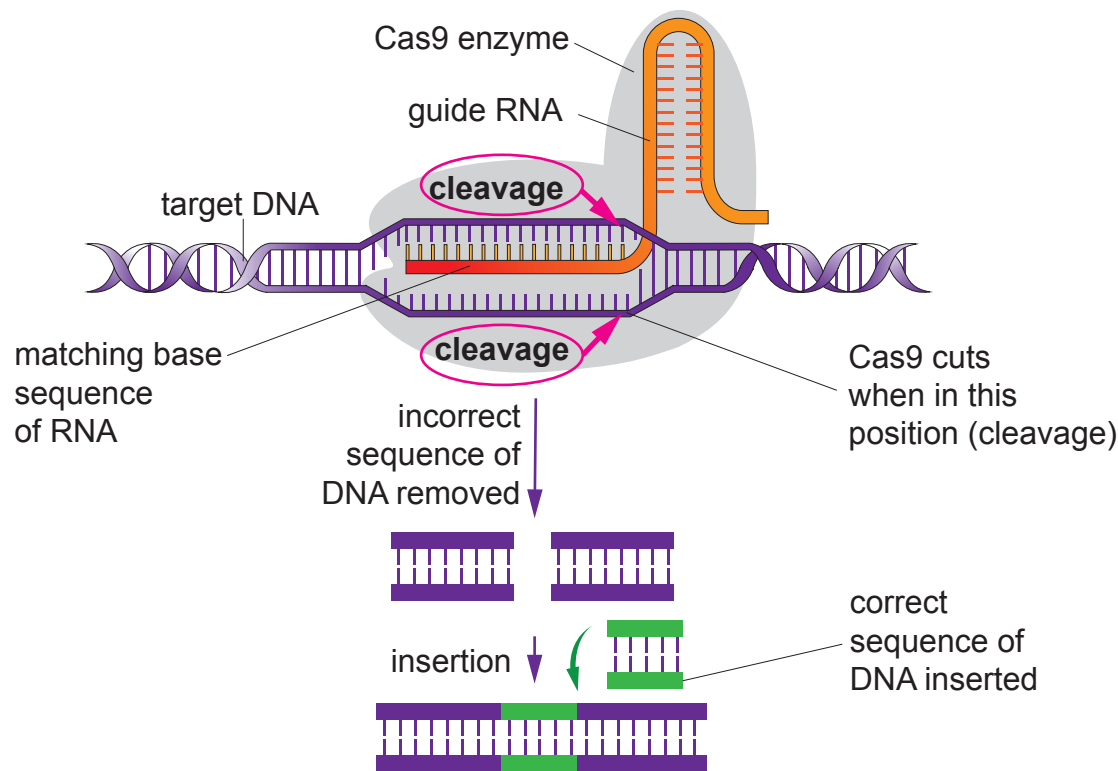
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The gene editing relies on the RNA guide recognising a target sequence of DNA so that the editing can be carried out on the correct section of DNA as shown in **Image 3.2**. This RNA is synthesised in laboratories from a template so that it matches the target sequence.

**Image 3.2**



(iii) Suggest how the results of the Human Genome Project has made the use of this gene editing technique possible. [2]

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